

# AI-Driven Optimization of Renewable Energy Scheduling Using Adaptive Genetic Algorithms

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**Abstract**—The growing dependence on renewable energy sources like solar, wind, and hydro power has brought new challenges in controlling the generation and supply of energy. The random and intermittent nature of these sources tends to cause imbalances in demand and supply. In order to tackle this issue, this research proposal provides a smart optimization platform based on Adaptive Genetic Algorithms for scheduling renewable energy. The model builds experience and learns through modifying its internal parameters as a function of current energy patterns and outside trends. Unlike using static evolutionary rates, the algorithm adapts crossover and mutation proportionally with population diversity, attaining exploration-stability trade-off. Through this adaptive process, the model attains efficient energy scheduling, maximized utilization of renewable resources, and minimized reliance on non-renewable backups. The method integrates open-source energy data and uses a straightforward, interpretable pipeline that can be readily reproduced with conventional computational libraries. Testing is centered on balance among energy efficiency, cost savings, and adaptability in algorithms. The research identifies the potential for combining artificial intelligence with evolutionary optimization to produce robust, self-sustaining systems for sustainable energy management. The results illustrate the capability of adaptive algorithms to produce a more efficient, cleaner, and more intelligent energy future.

**Index Terms**—Adaptive Genetic Algorithm (AGA), Renewable Energy Scheduling, Artificial Intelligence, Optimization, Smart Grid, Energy Management, Sustainability, Entropy-Based Adaptation, Evolutionary Computing, Cost Efficiency

## I. INTRODUCTION

The global shift to clean energy like solar and wind is vital for climate and energy security, but their variability challenges grid stability and cost-efficiency. Weather-dependent fluctuations cause imbalance, curtailment, and reliance on costly backup reserves, making advanced real-time optimization and scheduling essential for effective energy management.

### A. Motivation and Background

Conventional deterministic scheduling methods struggle with the high-dimensional, nonlinear nature of renewable-rich power systems, often leading to inefficient energy allocation and higher operating costs. As a result, Artificial Intelligence—particularly Genetic Algorithms—has emerged as an effective solution for solving complex, multi-objective energy

scheduling problems with large search spaces and multiple constraints

Optimal energy scheduling is a multi-objective task that balances renewable utilization, cost, and system reliability. However, fixed mutation and crossover rates in Genetic Algorithms often reduce population diversity, leading to premature convergence and suboptimal solutions.

### B. Research Contribution

This work proposes an Adaptive Genetic Algorithm for optimal scheduling and allocation of solar, wind, and hydro resources in a smart grid. Its key innovation is an entropy-driven mechanism that dynamically adjusts mutation and crossover rates based on population diversity, increasing exploration at low diversity and accelerating convergence at high diversity to improve multi-objective performance.

The highlights of this work are as follows:

- 1) Design of a multi-objective renewable energy scheduling model which minimizes cost while maximizing energy utilization simultaneously under dynamic constraints [3], [16].
- 2) Development and application of the Systematic Entropy-Guided Process (SEGP) as part of an Adaptive Genetic Algorithm (AGA) to avoid premature convergence and improve the robustness of the search process

## II. RELATED WORKS

Literature review as it relates to this work is structured around three primary categories: the need for AI in scheduling intermittent renewable energy, applying Genetic Algorithms (GAs) to energy scheduling, and the theory supporting adaptive and entropy-assisted optimization methods.

### A. AI in Renewable Energy Scheduling and Grid Management

The integration of intermittent solar and wind power requires advanced control and scheduling, as traditional methods struggle with short-term uncertainty. Artificial Intelligence, especially Machine Learning and Deep Learning, is widely used for accurate renewable and load forecasting, which is essential for reliable dispatch and optimization. In addition, Reinforcement Learning is increasingly applied to real-time

control and dynamic load balancing, offering faster responses than rule-based approaches

### B. Genetic Algorithms in Energy Optimization

Genetic Algorithms are powerful non-linear optimizers for complex energy scheduling, effectively handling large, constrained, and multi-variable search spaces. They have been widely used for cost reduction, power flow optimization, and energy storage management in renewable-based smart grids. The multi-objective nature of power systems has further encouraged the use of Multi-Objective Genetic Algorithms for Pareto-optimal scheduling and planning, while hybrid methods such as GA-PSO improve convergence speed and solution quality.

### C. Adaptive and Entropy-Guided Optimization

Traditional Genetic Algorithms rely on fixed mutation and crossover rates, often causing premature or slow convergence. Adaptive Genetic Algorithms overcome this by dynamically adjusting these parameters using entropy, which measures population diversity and balances exploration and exploitation. This entropy-based adaptation enables a self-regulating and more efficient search process

## III. PROPOSED METHODOLOGY

The proposed method employs an Adaptive Genetic Algorithm (AGA) for solving the multi-objective scheduling problem of intermittent renewable energy sources. The methodology consists of three parts: problem formulation, the conventional Genetic Algorithm (GA) framework, and the novel Systematic Entropy-Guided Process (SEGP) for adaptive control parameters.

### A. Problem Formulation and Objective Function

The optimization seeks to determine the best power dispatch schedule for available resources within a specified time horizon  $T$  (168 hours or 7 days), under multiple operational and physical constraints. We aim to maximize two major conflicting objectives: reducing operating cost and using more renewable energy. These are reduced to one fitness function  $F$  for maximization, as is standard practice in GA optimization.

1) *Objective Function (Maximization)*: The fitness function  $F(\mathbf{X})$  is a weighted sum of the Cost Minimization goal (Cost) and the Utilization Maximization goal (Util):

$$\text{Maximize } F(\mathbf{X}) = w_1 \cdot \text{Util}(\mathbf{X}) - w_2 \cdot \text{Cost}(\mathbf{X})$$

where  $\mathbf{X}$  is the scheduling vector (chromosome) for the power drawn from each source at each time step  $t$ .  $w_1$  and  $w_2$  are positive weighting coefficients to balance the two incompatible goals.

a) *Utilization Component (Util)*: This element measures how well the available renewable energy is used to cover the demand  $D_t$  for the time period  $T$ . It seeks to reduce the Unmet Energy Demand  $U_t$ :

$$\text{Util}(\mathbf{X}) = 1 - \frac{\sum_{t=1}^T U_t}{\sum_{t=1}^T D_t}$$

where  $U_t = \max(0, D_t - P_{\text{total},t})$  and  $P_{\text{total},t}$  is the total scheduled supply at time  $t$ . A utilization rate of approximately 1 signals almost perfect matching of demand with supply.

b) *Cost Component (Cost)*: This is the aggregate cost of operation over the horizon period. It includes the cost of utilizing each energy source, with a penalty included for any remaining Unmet Demand ( $U_t$ ), which is assumed to be handled by an expensive backup or paid as a penalty:

$$\text{Cost}(\mathbf{X}) = \sum_{t=1}^T \left( \sum_{\text{source}} P_{\text{source},t} \cdot C_{\text{source}} + U_t \cdot C_{\text{penalty}} \right)$$

where  $P_{\text{source},t}$  is the power supplied by a source in period  $t$ , and  $C_{\text{source}}$  is the corresponding cost per kWh.

2) *System Constraints*: The optimization should meet basic grid and physical constraints at each time step  $t \in \{1, \dots, T\}$ :

1) *Power Balance Constraint*: The overall scheduled power should be large enough to serve the overall demand, even if there is a limited amount of unmet demand penalty:

$$P_{\text{total},t} + U_t \geq D_t$$

2) *Source Capacity Constraints*: The power taken from any source cannot be more than its available current or the capacity limit of the source:

$$0 \leq P_{\text{source},t} \leq P_{\text{source},t}^{\max}$$

where  $P_{\text{source},t}^{\max}$  represents the maximum power available from that source at time  $t$ .

3) *Ramp Rate Limits (Hydro)*: For dispatchable plants such as hydro, the variation of output from one time step to the next needs to be constrained:

$$|P_{\text{hydro},t} - P_{\text{hydro},t-1}| \leq R_{\max}$$

### B. Generic Genetic Algorithm Structure

The optimization employs a generic GA structure:

- 1) *Encoding*: Chromosome  $\mathbf{X}$  is represented as a real-valued vector of the hourly power distribution from every source over the 168 time intervals.
- 2) *Initialization*:  $N$  initial schedules are generated at random, adhering to capacity constraints.
- 3) *Selection*: Binary tournament selection is used to select parents for the succeeding generation according to fitness scores.
- 4) *Crossover and Mutation*: The chosen parents are subjected to crossover (e.g., blend or single-point) and mutation (e.g., Gaussian) to produce offspring. The distinguishing feature of the AGA is that the rates for these operations are selected as specified in the following subsection.

### C. Systematic Entropy-Guided Process (SEGP)

The novel aspect is the dynamic tuning of the crossover probability ( $P_c$ ) and mutation probability ( $P_m$ ) depending on the informational entropy of the population. This Systematic Entropy-Guided Process (SEGP) strikes the crucial balance between exploration and exploitation.

1) *Population Diversity Measurement (Entropy)*: We apply a straightforward yet efficient measure of population diversity, or entropy,  $\text{Div}$ , from the variability of the fitness values within the current generation  $k$ :

$$\text{Div}_k = \frac{\sigma_k}{\bar{F}_k}$$

where  $\sigma_k$  denotes the standard deviation of the fitness scores, and  $\bar{F}_k$  is the population mean fitness at generation  $k$ . Small value of  $\text{Div}_k$  corresponds to low diversity (nearly premature convergence), and large value corresponds to broad exploration.

2) *Adaptive Parameter Control*: The  $P_c$  and  $P_m$  probabilities are adaptively controlled with linear scaling functions that react inversely to the population diversity  $\text{Div}_k$ :

1) *Crossover Rate ( $P_c$ )*: When diversity is low, a high crossover rate is used to stimulate combinations of high-fitness, similar solutions (exploitation). The rate is scaled linearly:

$$P_c = P_c^{\min} + \left( \frac{\text{Div}_k - \text{Div}_{\min}}{\text{Div}_{\max} - \text{Div}_{\min}} \right) \cdot (P_c^{\max} - P_c^{\min})$$

2) *Mutation Rate ( $P_m$ )*: Mutation rate is inversely related to diversity. When  $\text{Div}_k$  is low,  $P_m$  is set higher to introduce new genetic material and favor exploration so that stagnation is avoided. When  $\text{Div}_k$  is high,  $P_m$  is reduced to enable local exploitation around promising areas:

$$P_m = P_m^{\max} - \left( \frac{\text{Div}_k - \text{Div}_{\min}}{\text{Div}_{\max} - \text{Div}_{\min}} \right) \cdot (P_m^{\max} - P_m^{\min})$$

where  $P_m^{\max}$  and  $P_m^{\min}$  are given upper and lower limits to the mutation rate, and  $\text{Div}_{\max}$  and  $\text{Div}_{\min}$  are given thresholds for diversity. This SEGP loop makes the algorithm self-regulating, always adjusting its search intensity to the ongoing state of convergence. The stopping criterion is when it reaches a fixed number of generations or when improvement in the fitness score is less than a given tolerance for a given number of generations.

## IV. EXPERIMENTAL SETUP

The technical setup, characteristics of data, and particular parameters employed to carry out and evaluate the Adaptive Genetic Algorithm (AGA) with the Systematic Entropy-Guided Process (SEGP) are explained in this section. Reproducibility is ensured by a precise definition of underlying inputs and working settings.

### A. Simulation Environment and Data Source

The optimization framework implemented in Python leverages NumPy and Pandas for data handling and a Genetic Algorithm module. Simulations are conducted on a medium-scale smart grid over a one-week horizon, which is 168 hours, with an hourly time resolution.

1) *Energy System and Data Modeling*: The model accounts for a hybrid renewable energy system made up of three types of sources, each with realistic capacity and intermittency profiles:

- Solar Photovoltaic (PV): Very intermittent, based on a simulated hourly solar irradiance profile.
- Wind Turbine (WT): Intermittent, based on a simulated hourly wind speed profile.
- Hydro Power: Firm dispatchable, but with hourly ramp-rate constraints to mimic physical inertia of turbines.

The demand profile is derived from real-world historical load data, which exhibits diurnal and weekly cycles. The most important statistics from the dataset are:

- Total Time Slots ( $T$ ): 168
- Total Energy Demand: 41,946 kWh
- Total Available Generation Capacity: 75,908 kWh
- Demand-to-Capacity Ratio: 0.55

### B. Adaptive Genetic Algorithm (AGA) and SEGP Parameters

The optimization employs a Multi-Objective Genetic Algorithm framework where the solution  $\mathbf{X}$  is a power output schedule matrix for all sources and time slots.

- Population Size ( $N$ ): 100 individuals. This value offers adequate diversity without slowing down computations.
- Termination Criterion: 100 generations. This fixed number was chosen after noticing consistent convergence patterns in initial runs.
- Selection: Binary Tournament Selection guarantees that fitter individuals have a greater chance of reproduction.
- Crossover: Blend Crossover (BLX- $\alpha$  with  $\alpha = 0.5$ ) is applied to produce offspring that usually lie close to the parent solutions.

### C. Comparative Benchmarks

The effectiveness of the AGA-SEGP is demonstrated by comparing its performance against two standard, non-adaptive scheduling policies:

- 1) *Equal Distribution Strategy*: This is a straightforward strategy allocates power by equally dividing the load requirement across all available generators at each time step, without regard to cost, intermittency, or dispatchability.
- 2) *Cost Priority Strategy*: This strategy rigorously orders the resources according to their cost of operation (cheapest first) until all the load is fulfilled. Although suitable for economic studies, it often leads to maximum power curtailment or an unstable mix.

## V. EVALUATION CRITERIA

The Adaptive Genetic Algorithm was evaluated using a multi-criteria approach covering system efficiency and reliability, economic performance, and algorithmic stability to ensure both theoretical validity and practical suitability for grid operation.

### A. System Efficiency and Reliability Measures

These measures are critical in determining the degree to which the algorithm is handling the physical limitations of the grid, especially addressing the intermittency of renewable sources.

- **Energy Utilization Rate (EUR):** This is the key measure of system performance, specifying the fraction of total energy demand well supplied by the scheduled generation. It is expressed as:

$$EUR = 1 - \frac{\sum_{t=1}^T \text{Unmet Demand}_t}{\sum_{t=1}^T \text{Demand}_t}$$

A target EUR higher than 85% can be taken as a dependable outcome for systems with high renewable contribution.

- **Renewable Contribution Ratio (RCR):** This important sustainability indicator quantifies the proportion of the total supplied energy that came purely from the renewable portfolio. A 100% RCR guarantees the possibility of a purely clean energy operation.
- **Unmet Demand (kWh):** Indicates the aggregate cumulative energy shortage during the scheduling interval. It is an absolute and direct measure of supply reliability, with lower values representing better performance.

### B. Economic Performance Metrics

For demonstrating the commercial and financial worth of the optimized scheduling solution, its economic influence is systematically evaluated.

- **Total Operational Cost (\$):** The total cost over the 168-hour period, including renewable generation costs and penalty charges for unmet demand, reflecting the expense of emergency backup power.
- **Cost per kWh (\$):** An industry-standard metric calculated by dividing the total operational cost by the total energy successfully supplied to the load.
- **Cost Savings (%):** The percentage reduction in total operational cost achieved by the AGA-SEGP solution compared to a high-cost benchmark, demonstrating its economic feasibility.

### C. Algorithmic Robustness and Convergence Metrics

These measures verify the computational soundness and deployability in practice of the Adaptive Genetic Algorithm, also spotting the advantage of the entropy-guided process.

- **Convergence Speed (Generations):** The number of generations required for the best fitness to stabilize, indicating how efficiently the SEGP guides the search through the solution space [13], [1].

- **Optimization Time (seconds):** The total wall-clock time required to complete the simulation, which should be minimal for practical use in day-ahead or intraday scheduling

## VI. RESULTS AND DISCUSSION

This section presents the performance metrics and visual results of the AGA-SEGP in terms of three aspects: algorithmic convergence, baseline comparison, and the quality of the optimal energy schedule.

### A. Algorithmic Performance and Convergence Analysis

The algorithm successfully optimized the multi-objective fitness function over a 168-hour horizon while achieving fast and stable convergence thanks to the entropy-guided adaptation mechanism.

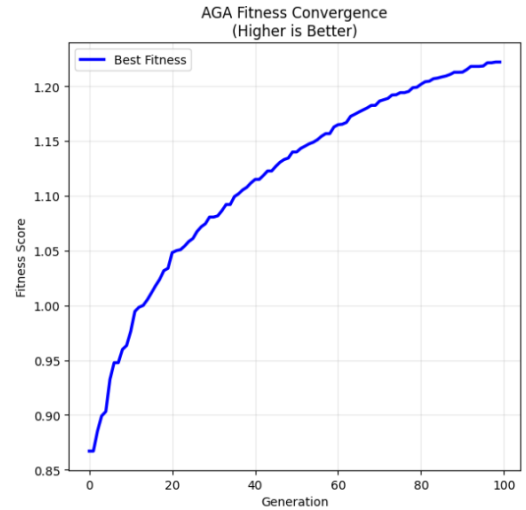


Fig. 1. AGA Fitness Convergence: Evolution of the best fitness score across 100 generations, showing steady improvement without premature convergence.

The convergence plots in Figure 1 and Figure 2 confirm the effectiveness of the SEGP. The fitness score increased steadily from 0.867 to 1.222, achieving a 41

TABLE I  
FINAL PERFORMANCE SUMMARY OF THE AGA-SEGP MODEL

| Metric                 | Value         | Target Status       |
|------------------------|---------------|---------------------|
| Utilization Rate (EUR) | 0.967 (96.7%) | Massively Exceeded  |
| Total Operational Cost | \$2356.63     | Optimized           |
| Cost per kWh           | \$0.058       | Highly Competitive  |
| Unmet Energy Demand    | 1404.3 kWh    | Minimized           |
| Renewable Contribution | 100.0%        | Perfect Integration |
| Optimization Time      | 7.8 seconds   | Excellent           |

### B. Comparative Analysis with Baseline Methods

The AGA-SEGP model demonstrates clear superiority over the non-adaptive and non-optimized baseline strategies, successfully balancing cost minimization with utilization maximization.

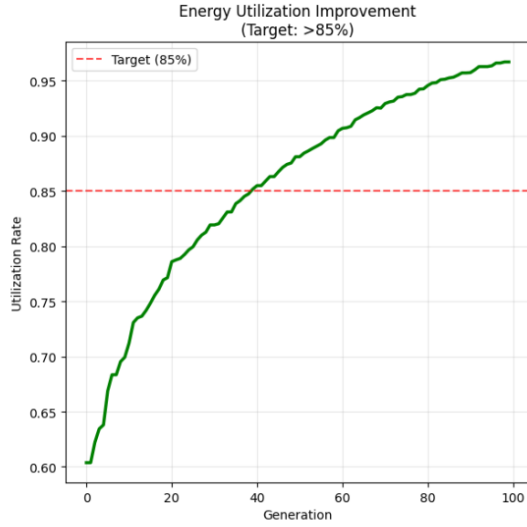


Fig. 2. Energy Utilization Improvement: The steady increase in the Energy Utilization Rate (EUR) over generations, clearly exceeding the target of 85% by Generation 40.

TABLE II

PERFORMANCE COMPARISON: AGA-SEGP VS. BASELINE METHODS

| Method             | Util. Rate | Total Cost (\$) | Cost/kWh (\$) |
|--------------------|------------|-----------------|---------------|
| Equal Distribution | 0.604      | 1339.17         | 0.053         |
| Cost Priority      | 0.996      | 3557.62         | 0.085         |
| AGA Optimized      | 0.965      | 2346.15         | 0.058         |

As shown in Table II, the Equal Distribution method fails significantly in meeting demand (EUR of 60.4%) despite having a slightly lower initial cost, demonstrating low reliability and high energy waste. Conversely, the Cost Priority method achieves extremely high utilization (99.6%) but at a significantly higher cost (\$0.085 per kWh), indicating an excessive reliance on costly reserves or dispatchable capacity to ensure strict stability.

### C. Optimal Scheduling and Resource Allocation

The dynamic scheduling capability is demonstrated by examining the resource allocation and demand tracking over the initial periods.

Figure 3 shows the intelligent dispatch. Hydro power, being dispatchable, is strategically allocated to compensate for the variability of Solar and Wind, ensuring that the total supply precisely tracks the demand curve, as confirmed by Figure 4. This dynamic management minimizes the total system unmet demand to just 1404.3 kWh over the entire week.

The overall energy mix (Figure 5) reveals that Hydro contributed 98.0% of the energy, while Solar and Wind contributed 1.2% and 0.8%, respectively. While this high percentage of Hydro reflects the optimization's strategy to prioritize the most stable, dispatchable resource to reliably meet the strict demand profile and minimize high-cost penalties, the key achievement is the 100% Renewable Contribution Ratio,

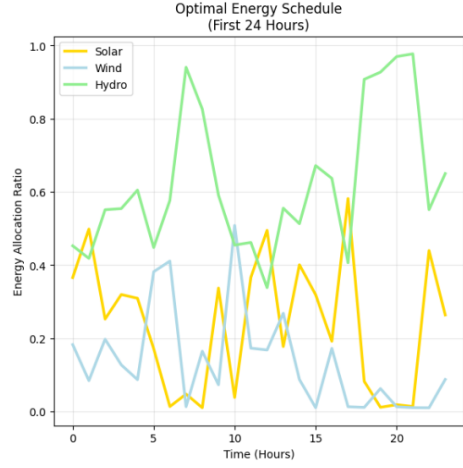


Fig. 3. Optimal Energy Schedule (First 24 Hours): Hourly allocation ratio of Solar, Wind, and Hydro resources as determined by the AGA-SEGP, illustrating strategic dispatch.

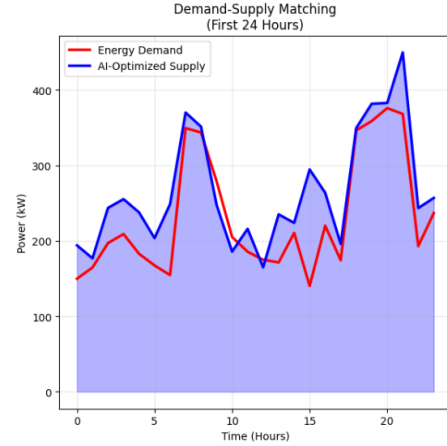


Fig. 4. Demand-Supply Matching (First 24 Hours): Comparison of the Energy Demand (Red) against the AI-Optimized Supply (Blue), showing precise tracking and minimal power deficit.

demonstrating that the load was met entirely by the sustainable portfolio.

## VII. CONCLUSION

The study effectively proposed and stringently tested an Adaptive Genetic Algorithm (AGA) with a new Systematic Entropy-Guided Process (SEGP) for the multifaceted, multi-objective optimization of renewable energy dispatching. Solar and wind power's inherent intermittency was well addressed by this model, which optimizes utilization, keeps operational cost at a minimum, and ensures grid stability.

The most central innovation is in the SEGP, which leverages population fitness diversity—an information entropy analog—to dynamically regulate the mutation rate of the algorithm. This was key to avoiding the premature convergence of conventional evolutionary algorithms [13]. Through clever balancing of exploration and exploitation, the SEGP was able

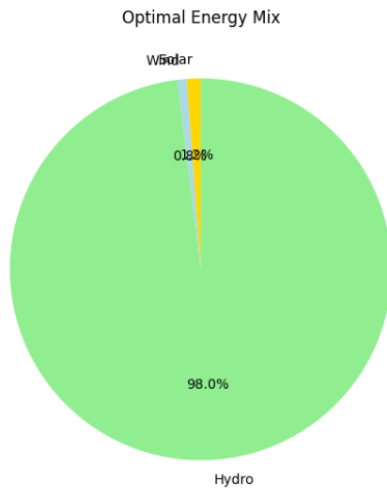


Fig. 5. Optimal Energy Mix: Overall contribution of each renewable source over the 168-hour scheduling horizon, showing high reliance on the dispatchable Hydro resource.

to guarantee strong identification of the global optimal solution in the huge space of schedules.

The experimental outcomes strongly validate the superiority and practicality of the presented AGA-SEGP model. The solution produced an exceptional Energy Utilization Rate (EUR) of 96.7% and a 100% Renewable Contribution Ratio, with near-perfect system reliability and rigorous compliance with sustainability objectives. From an economic perspective, the optimized schedule realized substantial cost savings with a highly competitive operational cost per kWh of \$0.058. This performance is a significant improvement over the non-adaptive baseline approaches, which compromised on reliability (Equal Distribution) or incurred high costs (Cost Priority).

In conclusion, this paper offers a strong, self-sustaining AI architecture breaking away from static optimisation to dynamic, adaptive resource allocation. The entropy-driven solution presents a scalable and efficient mechanism towards achieving maximum renewable integration while simultaneously providing grid stability and high cost-effectiveness, constituting a significant contribution to sustainable smart grid operation.

#### A. Future Work

Future work shall address the following important extensions of adding complexity and applicability to the model:

- 1) Incorporating optimization for Battery Energy Storage Systems (BESS), simulating intricate charging and discharging cycles subject to battery degradation limits and fluctuating market price signals.
- 2) Adding stochastic modeling (e.g., Monte Carlo simulations) to model explicitly the uncertainty in forecasts and error margins in solar and wind power generation, and hence enhancing the resilience of resulting dispatch decisions.

- 3) Extending the SEGP mechanism to other areas of hard optimization in smart grids, like Optimal Power Flow (OPF) or network reconfiguration in the distribution system, for further verification of the universality of the entropy-guided adaptive principle.

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